CS5224 CLOUD COMPUTING · GROUP 20

# Ansera

An AI admin on **WhatsApp** for Indonesian micro, small & medium businesses




SaaS on Google Cloud | Singapore region | No-code setup

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PROBLEM STATEMENT

# Indonesian businesses run on WhatsApp, and answer it all by hand

**90%+** [1]

of Indonesian internet users use WhatsApp



**8 / 8** our survey

businesses we surveyed rely on it to talk to customers



**60%** [2]

of customers expect a reply within **10 minutes**



🌙

**7 of 8**

miss messages after hours, and lose customers

⏳

**3–5+ hours a day**

spent just replying to chats

💸

**Up to Rp3M a month**

to hire an admin, with high turnover

🌙 02:14 **9**

**Rina** Cash on delivery OK?

**Tono** Is it in stock?

**Lina** I'd like 3 boxes 🙏

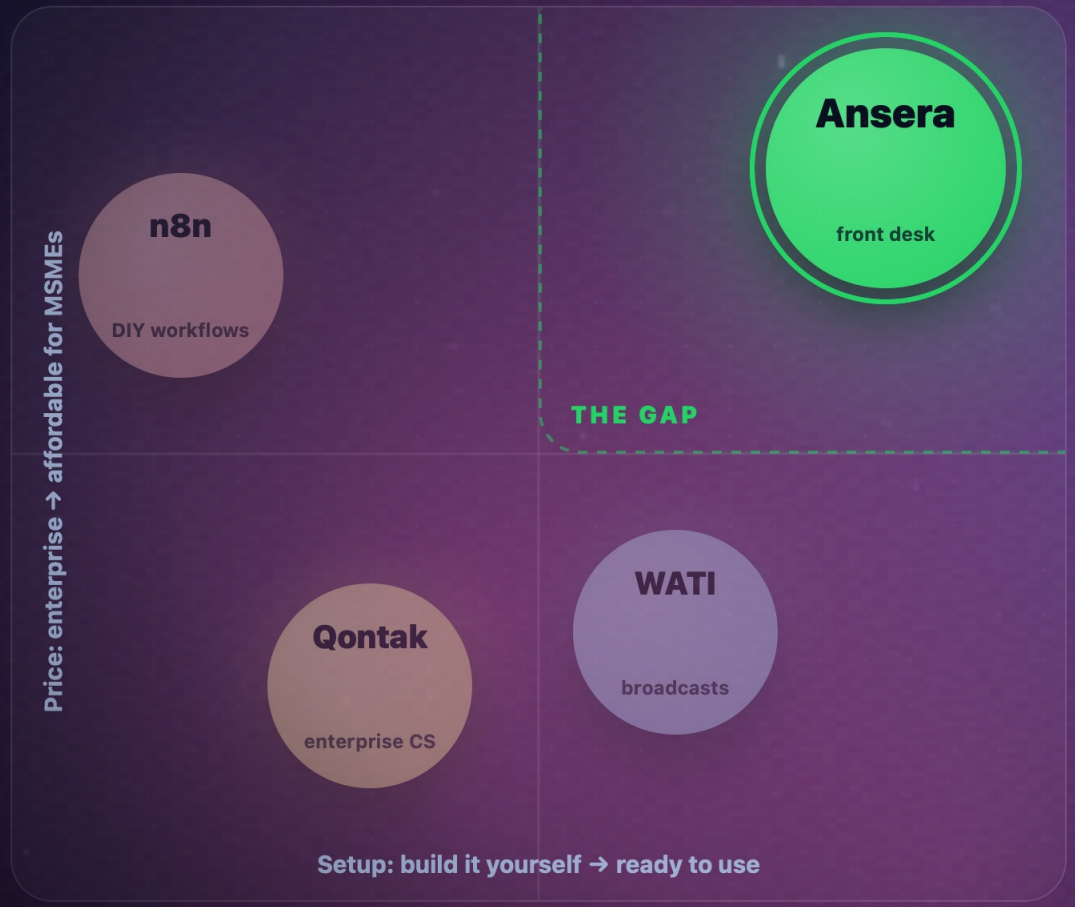
**Agus** Hello? 😕

**Maya** Still open?

[1] We Are Social & DataReportal, Digital 2026: Indonesia · [2] crm.id, "Kesalahan UMKM" (industry blog) · survey of 8 Indonesian MSME owners (our preliminary report)

COMPETITIVE ANALYSIS

# Too big, too narrow, or build it yourself




**Mekari Qontak**
All-in-one customer service for large companies: comprehensive features, but expensive, with heavy onboarding

**WATI**
Mainly a WhatsApp broadcast tool for online stores

**n8n**
General automation for developers: you build every workflow yourself

**The gap: affordable and easy**
A lightweight front desk for SMEs, with no building: it understands the owner's own product info and replies **24/7**

SOLUTION

# Ansera: an AI admin on WhatsApp, your own front desk

## A customisable front-desk flow

click a block to configure it

👋 Welcome → ❓ Answer from FAQs → 🛒 Order / booking → 🧾 Invoice + pay → ⏰ Reminders

⚙️ Welcome block: welcome message | reply style | details to collect: name, email, players

### Answers questions 24/7

From the business's own product info: products, services, FAQs

### Orders & bookings

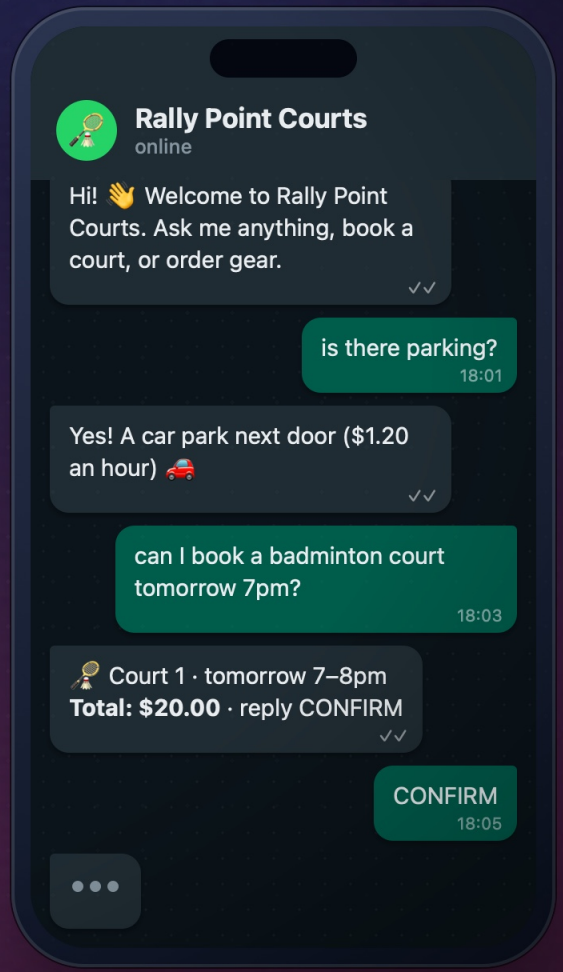
Taken right in the chat

### Payment with invoices

Guides the customer through paying

### Reminders

For unpaid orders and upcoming bookings



ARCHITECTURE · GOOGLE CLOUD, SINGAPORE

# A SaaS built on PaaS, in three planes




SaaS Ansera

PaaS Cloud Run · Pub/Sub · Cloud SQL · Vertex AI

IaaS Google Cloud, asia-southeast1

- ✓ Control / client plane
- ✓ Message plane
- ✓ Payment plane
- 4 Shared platform

USER FLOW · THE CUSTOMER'S POINT OF VIEW

# ① A customer messages the shop, through to a payment link




- ✓ Signed webhook in
- ✓ Signature verified, published
- ✓ Kept in the queue
- ✓ Cloud SQL → LLM reply
- ✓ Reply with the shop's token
- 6 CONFIRM → invoice + link

USER FLOW · THE CUSTOMER'S POINT OF VIEW

# ② Paid: receipt and reminders




- ✓ Pay on Xendit's page
- ✓ Verified, paid · receipt by email + WhatsApp
- 3 Hourly reminders

EVENT PIPELINE · A CLOSER LOOK AT THE QUEUES

# Answer Meta fast, handle each message once




### ⚡ 200 in ~100 ms

The ingester only checks the signature and publishes. If the publish fails: `503`, and Meta retries

### 🪪 Claimed by its WhatsApp ID

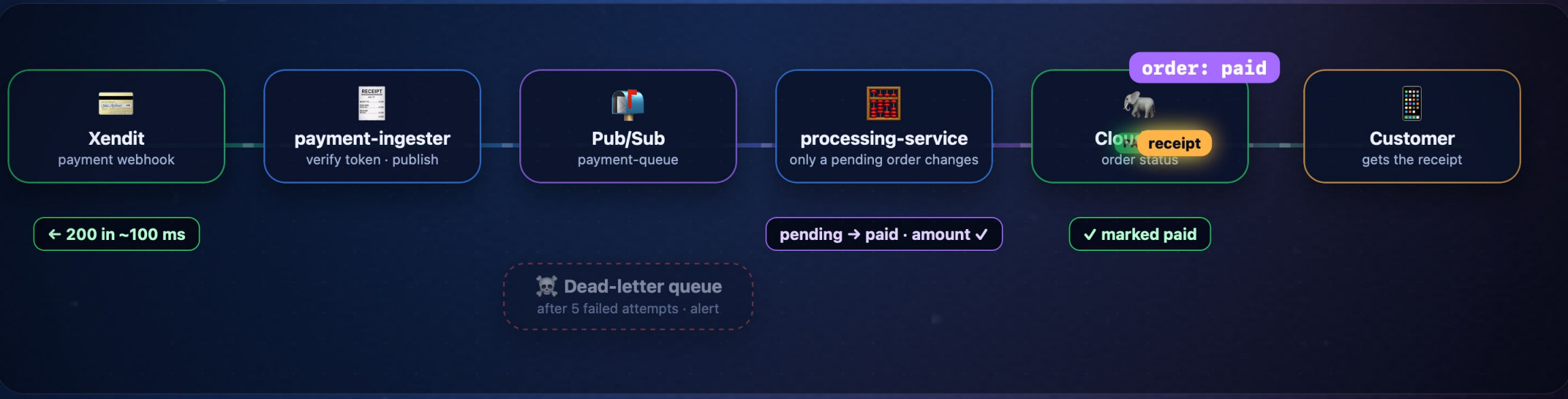
Pub/Sub delivers at least once: each message is claimed by its unique ID, so it's processed once

### 🧵 A burst gets one reply

No ordering key, on purpose: each new message bumps the chat's version, and a turn saves its reply only if the version is unchanged, so one reply answers them all

EVENT PIPELINE · A CLOSER LOOK AT THE QUEUES

# Failures are expected, so we designed for them




### 🔢 One turn, one transaction

State, order and reply commit together, in one transaction, under a row lock

### ☠️ Retry, then dead-letter

Poison messages back off from 30 s to 10 min; after 5 failed attempts, the dead-letter queue

### 💳 Payments too

Only a pending order can change, so a repeated Xendit event is a no-op, and a late expiry never cancels a paid order

LLM ENGINE · GEMINI ON VERTEX AI

# The system decides, the LLM talks

**CUSTOMER** can I get 2 Astrox 99 Pro, and where's my last order?

## 🧭 Router

one LLM call · JSON out

```
{ "intents": ["order", "order_status"] }
```

`order` `booking` `question` `order_status`

## 🛒 Orders & bookings

`state machine` → LLM extracts the details → live stock & free slots

## 📦 Order status

this customer's orders only → `unknown numbers blocked`

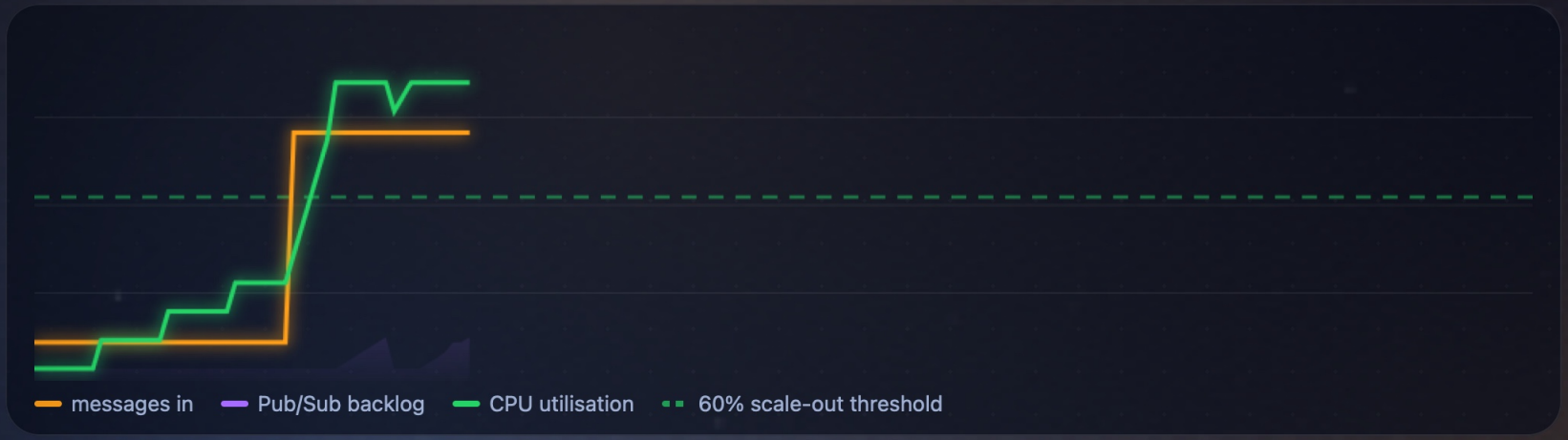
## ❓ Questions

read-only, tenant-scoped tools → products · services · FAQs → unknown → owner

LLM call · Code check · Database · Fixed template

SCALING

# Cloud Run scales out, Pub/Sub absorbs the burst




container #1

container #2

TRAFFIC
9 msg/s
WhatsApp chats + Xendit webhooks

CPU UTILISATION
100%

INSTANCES (CAP 3)
2 / 3

BACKLOG IN PUB/SUB
6

CPU over 60%: Cloud Run adds a container

Containers on Cloud Run · Horizontal autoscaling at CPU > 60% · Capped at 3 per service (demo) · Pub/Sub buffer · asynchronous

SECURITY

# Private by default, one guarded front door



MULTI-TENANCY

# One platform for every business, each one walled off

## Shared by every tenant

Rally Point · Kopi & Co. · Lumen & Lens

↓ application-level isolation ↓

one set of backend containers · one PostgreSQL database

## Authenticated APIs

request `Authorization: Bearer eyJhbGciOi…`

↓ JWT verified

`business_id = 5` every request is tied to its business's operation

## Business-ID isolation

5 Wei Ling: court tomorrow 7pm?

`GET /api/chats/ (business 1)` 403

Each WhatsApp number belongs to **one business only**: no duplicates across businesses

## Unique within each business

```
UNIQUE (business_id, product_name)
```

✓ 5 · Yonex Astrox 99 Pro

~~× 5 · Yonex Astrox 99 Pro~~ duplicate

No ambiguity in what's on offer, or in the orders that follow

MULTI-TENANCY

# The data layer: fast, consistent, and safe

## ⚡ Indexed, and strongly consistent

indexes on `order_status` `chat_session_id` `business_id`

**Consistency ✓** OVER ~~Availability~~

Never stale data: accurate stock counts and payment values

## 🔐 Concurrency: two kinds of lock

| | |
|---|---|
| 💳 **Payments & stock** | pessimistic: the row is locked first `SELECT … FOR UPDATE` |
| 💬 **Customer chats** | optimistic: async LLM calls, chat versions `chat v7 → v8 · a stale turn reruns` |

## 🗡️ SQL injection, mitigated twice

`name = "x' OR 1=1 --"` × just text to the ORM

- 🧱 well-defined ORMs
- 🧽 sanitised serializers
- 🛡️ Cloud Armor on the load balancer

## 🗃️ One bucket, grouped by tenant

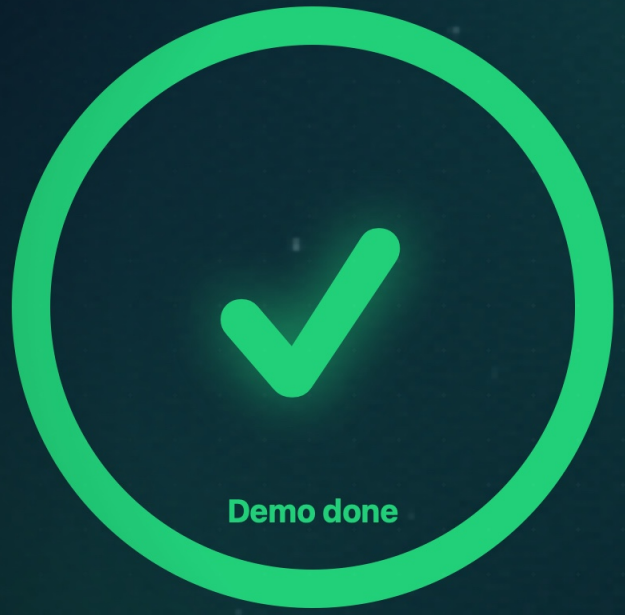
```
invoices/5/R-0008.pdf
receipts/5/R-0008.pdf
```

🔗 Customers get a **7-day signed URL**, never bucket access

**Standard** → older files → **Archive** lifecycle policy, lower storage cost

LIVE DEMO · 3 MINUTES

# Rally Point Courts: an owner and a customer, live




- 🧩 **The owner sets up**
  Dashboard · workflow settings · products, services, FAQs
- 💬 **A customer asks**
  "is there parking?" → FAQ · "can I bring my own coach?" → owner
- 🏸 **Book & order**
  Court tomorrow 7pm · 2 rackets but 1 left → never oversells · CONFIRM
- 💳 **Pay**
  Invoice email · Xendit test page → receipt on WhatsApp + email
- 📊 **Back to the owner**
  "Paid, awaiting delivery" · stock updated · answer → FAQ · reminders

EVALUATION

# Functional, technical, and tested with users

## ① FUNCTIONAL

**1,371**

tests across **7 services**

Authentication · Catalogue · Chat · Payment · Reminders · Dashboard

No failures · No errors

## ② TECHNICAL

| | |
|---|---|
| Webhook p95 | 196 ms |
| Sandbox reply latency p95 | 3.26 s |
| HTTP burst<br>read-only | 455 / 455 ✓ |
| Fault & duplicate scenarios | 12 / 12 ✓ |
| Dead-letter handling<br>failed message reached the DLQ | 78.19 s |
| Tenant isolation<br>cross-tenant requests | 403 Forbidden |

## ③ USER TESTING

🧁 **5 participants** as home-bakery owners: set up the shop, added products & FAQs, chatted as customers, paid, booked

| | |
|---|---|
| **Intuitiveness** | 4.0/5 |
| **Usability** | 4.3/5 |
| **Usefulness** | 4.3/5 |

⚠️ Main issue: navigation. Products, services and FAQs all sit under one **Catalog** section

The HTTP burst test was read-only

BUSINESS MODEL

# B2B SaaS, from Rp149,000 a month

## Starter

**Rp149,000** (S$10.57) / month

**300** interactions

1 WhatsApp number, no-code workflow builder, orders and bookings, invoices and receipts, payment and reminder follow-ups

## Growth

**Rp499,000** (S$35.39) / month

**1,500** interactions

Starter, plus 3 staff logins*, analytics dashboard*

## Business

**Rp1,499,000** (S$106.31) / month

**4,000** interactions

Growth, plus 10 logins*, 3 WhatsApp numbers*, priority support*

💡 Starter is about **4–7%** of a dedicated admin's Rp2–3.5M (S$142–248) monthly salary [6], and Ansera replies at any hour

YEAR 1: 100 BUSINESSES

**60 · 30 · 10**

Starter · Growth · Business

MONTHLY REVENUE

**Rp38.9M**

(S$2,760) · 61,000 interactions

MONTHLY COST

**Rp29.8M**

(S$2,111) · Rp21.0M variable, Rp8.8M fixed

MONTHLY PROFIT

**Rp9.1M**

(S$649) · 23% margin

BREAK-EVEN

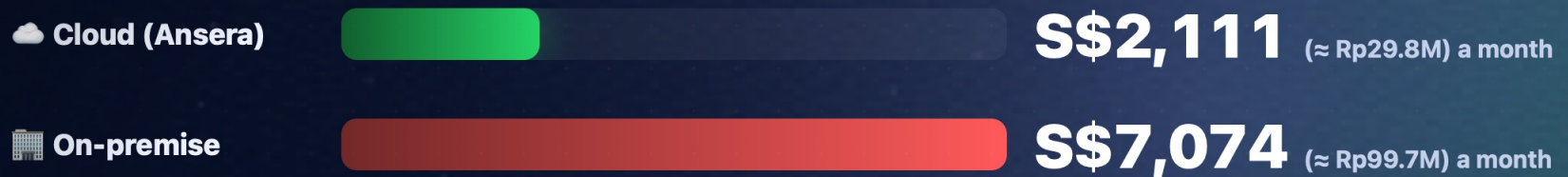
**≈ 50**

businesses

Interaction = all of one customer's messages within 24 hours  Prices exclude VAT  *Planned for future versions  Meta messaging and Xendit fees are paid by each business [10, 11]

ESTIMATED COST COMPARISON

# On-premise would cost about 3.3× more




**≈ 3.3×**
more expensive on-premise
3-year TCO: S$254.7K vs S$76.0K

## Cloud: how we got S$2,111

| COST COMPONENT | MONTHLY COST |
|---|---|
| Cloud Run (5 services, always on) | S$200 (Rp2.8M) |
| Cloud SQL (PostgreSQL) | S$97 (Rp1.4M) |
| Load balancer and Cloud Armor | S$38 (Rp0.5M) |
| Storage, Pub/Sub, logging, network | S$35 (Rp0.5M) |
| Gemini AI inference and request compute | S$1,108 (Rp15.6M) |
| Operations / on-call | S$255 (Rp3.6M) |
| Email (Brevo) | S$59 (Rp0.8M) |
| Support | S$319 (Rp4.5M) |
| **Total monthly cost** | **S$2,111** (Rp29.8M) |
| Three-year TCO | S$76.0K (Rp1.07B) |

## On-premise: how we got S$7,074

| COST COMPONENT | MONTHLY COST |
|---|---|
| Hardware depreciation (2 app servers, 2 GPU servers, network, storage, US$65,000 over 36 months) | S$2,318 (Rp32.7M) |
| Colocation and power (3.9 kW at US$450/kW) | S$2,253 (Rp31.8M) |
| Operations / staff | S$766 (Rp10.8M) |
| Hardware support and spares (8% a year) | S$556 (Rp7.8M) |
| Bandwidth and DDoS protection | S$513 (Rp7.2M) |
| Software licences | S$160 (Rp2.3M) |
| Backup and disaster recovery copy | S$128 (Rp1.8M) |
| Email (Brevo) | S$59 (Rp0.8M) |
| Support | S$319 (Rp4.5M) |
| **Total monthly cost** | **S$7,074** (Rp99.7M) |
| Three-year TCO | S$254.7K (Rp3.59B) |

3-year horizon, serving 100 businesses  Monthly estimates, modelled, not billed · exchange rates: Rp14,100 / SGD, Rp18,100 / USD

LIMITATIONS

# What it can't do yet, and how we'd fix it

## Text only, one number

- Text messages only, through WhatsApp
- One WhatsApp number per business

**THE FIX**
Multimodal input, like images

## Very large catalogues

- The whole catalogue has to fit in the model's input

**THE FIX**
RAG: retrieve only the items a question needs

## One region, one cloud

- Singapore only: recovering from a regional outage is manual
- Depends on GCP: some vendor lock-in

**THE FIX**
A backup deployment in Jakarta, with cross-region traffic routing and failover

FUTURE WORK

# What's next, on top of the fixes

### 🙋 Human takeover

The owner steps into a conversation whenever the AI can't handle the situation

### 🧭 Guided setup

Helps first-time users get started

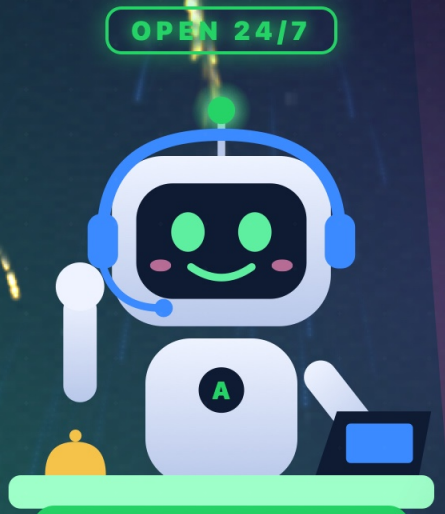
### 🔄 Stock sync

With marketplaces and physical warehouses

THANK YOU

CONCLUSION

# Routine customer chats, handled 24/7




💬

**Answers, sells, gets paid**

An AI admin on WhatsApp, set up without code

☁️

**SaaS on Google Cloud**

Scales out, and designed for failure

💸

**≈ 3.3× cheaper**

than running it on-premise (estimated)

## Thank you! 🙌